

GE Healthcare

SmartStep
Discovery 600 Family
Discovery CT750 HD

Operating Documentation



5323334-1EN
Revision: 1

Effectivity

The information in this manual applies to the following CT and PET/CT Scanners:

Discovery CT750 HD

Discovery 600 Family

LEGAL NOTES

TRADEMARKS

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IMPORTANT PRECAUTIONS

LANGUAGE

Предупреждение

(BG)

- ТОВА УПЪТВАНЕ ЗА РАБОТА Е НАЛИЧНО САМО НА АНГЛИЙСКИ ЕЗИК.
- АКО ДОСТАВЧИКЪТ НА УСЛУГАТА НА КЛИЕНТА ИЗИСКА ЕЗИК, РАЗЛИЧЕН ОТ АНГЛИЙСКИ, ЗАДЪЛЖЕНИЕ НА КЛИЕНТА Е ДА ОСИГУРИ ПРЕВОД.
- НЕ ИЗПОЛЗВАЙТЕ ОБОРУДВАНЕТО ПРЕДИ ДА СТЕ СЕ КОНСУЛТИРАЛИ И РАЗБРАЛИ УПЪТВАНЕТО ЗА РАБОТА.
- НЕСПАЗВАНЕТО НА ТОВА ПРЕДУПРЕЖДЕНИЕ МОЖЕ ДА ДОВЕДЕ ДО НАРАНЯВАНЕ НА ДОСТАВЧИКА НА УСЛУГАТА, ОПЕРАТОРА ИЛИ ПАЦИЕНТ В РЕЗУЛТАТ НА ТОКОВ УДАР ИЛИ МЕХАНИЧНА ИЛИ ДРУГА ОПАСНОСТ.

警告

(ZH-CN)

- 本维修手册仅提供英文版本。
- 如果维修服务提供商需要非英文版本，客户需自行提供翻译服务。
- 未详细阅读和完全理解本维修手册之前，不得进行维修。
- 忽略本警告可能对维修人员，操作员或患者造成触电、机械伤害或其他形式的伤害。

VÝSTRAHA

(CS)

- Tento provozní návod existuje pouze v anglickém jazyce.
- V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka.
- Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah.
- V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.

ADVARSEL

(DA)

- Denne servicemanual findes kun på engelsk.
- Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse.
- Forsøg ikke at servicere udstyret medmindre denne servicemanual har været konsulteret og er forstået.
- Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk, mekanisk eller anden fare for teknikeren, operatøren eller patienten.

WAARSCHUWING

(NL)

- Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar.
- Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan.
- Probeer de apparatuur niet te onderhouden voordat deze onderhoudshandleiding werd geraadpleegd en begrepen is.
- Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

WARNING

(EN)

- This Service Manual is available in English only.
- If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services.
- Do not attempt to service the equipment unless this service manual has been consulted and is understood.
- Failure to heed this warning may result in injury to the service provider, operator, or patient, from electric shock or from mechanical or other hazards.

HOIATUS

(ET)

- Käesolev teenindusjuhend on saadaval ainult inglise keeles.
- Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest.
- Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist.
- Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.

VAROITUS

(FI)

- Tämä huolto-ohje on saatavilla vain englanniksi.
- Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla.
- Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen.
- Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.

ATTENTION

(FR)

- Ce manuel de service n'est disponible qu'en anglais.
- Si le technicien du client a besoin de ce manuel dans une autre langue que l'anglais, c'est au client qu'il incombe de le faire traduire.
- Ne pas tenter d'intervenir sur les équipements tant que le manuel service n'a pas été consulté et compris.
- Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.

WARNUNG

(DE)

- Diese Serviceanleitung existiert nur in Englischer Sprache.
- Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine Entsprechende Übersetzung zu sorgen.
- Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben.
- Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, Mechanische oder Sonstige Gefahren kommen.

ΠΡΟΕΙΔΟΠΟΙΗΣΗ

(EL)

- Το παρόν εγχειρίδιο σέρβις διατίθεται στα αγγλικά μόνο.
- Εάν το άτομο παροχής σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει υπηρεσίες μετάφρασης.
- Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό εκτός εάν έχετε συμβουλευτεί και έχετε κατανοήσει το παρόν εγχειρίδιο σέρβις.
- Εάν δε λάβετε υπόψη την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στο άτομο παροχής σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.

FIGYELMEZTETÉS

(HU)

- Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el.
- Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészítése.
- Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték.
- Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.

AÐVÖRUN

(IS)

- Þessi þjónustuhandbók er eingöngu áanleg á ensku.
- Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálþjónustu.
- Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin.
- Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.

AVVERTENZA

(IT)

- Il presente manuale di manutenzione è disponibile soltanto in inglese.
- Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione.
- Si proceda alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto.
- Il non rispetto della presente avvertenza potrebbe far compiere operazioni da cui derivino lesioni all'addetto, alla manutenzione, all'utilizzatore ed al paziente per folgorazione elettrica, per urti meccanici od altri rischi.

警告

(JA)

- このサービスマニュアルには英語版しかありません。
- サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。
- この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

경고

(KO)

- 본 서비스 지침서는 영어로만 이용하실 수 있습니다 .
- 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우 , 번역 서비스를 제공하는 것은 고객의 책임입니다 .
- 본 서비스 지침서를 참고했고 이해하지 않는 한은 해당 장비를 수리하려고 시도하지 마십시오 .
- 이 경고에 유의하지 않으면 전기 쇼크 , 기계상의 혹은 다른 위험으로부터 서비스 제공자 , 운영자 혹은 환자에게 위해를 가할 수 있습니다 .

BRDINJUMS

(LV)

- Šī apkalpes rokasgrāmata ir pieejama tikai angļu valodā.
- Ja klienta apkalpes sniedzējam nepieciešama informācija citā valodā, nevis angļu, klienta pienākums ir nodrošināt tulkošanu.
- Neveiciet aprīkojuma apkalpi bez apkalpes rokasgrāmatas izlasīšanas un saprašanas.
- Šī brīdinājuma neievērošana var radīt elektriskās strāvas trieciena, mehānisku vai citu risku izraisītu traumu apkalpes sniedzējam, operatoram vai pacientam.

ĮSPĖJIMAS

(LT)

- Šis eksploatavimo vadovas yra prieinamas tik anglų kalba.
- Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, numatyti vertimo paslaugas yra kliento atsakomybė.
- Nemėginkite atlikti įrangos techninės priežiūros, nebent atsižvelgėte į šį eksploatavimo vadovą ir jį supratote.
- Jei neatkreipsite dėmesio į šį perspėjimą, galimi sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų paslaugų tiekėjui, operatoriui ar pacientui.

ADVARSEL

(NO)

- Denne servicehåndboken finnes bare på engelsk.
- Hvis kundens serviceleverandør trenger et annet språk, er det kundens ansvar å sørge for oversettelse.
- Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått.
- Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.

OSTRZEŻENIE

(PL)

- Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim.
- Jeśli dostawca usług klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta.
- Nie próbować serwisować wyposażenia bez zapoznania się i zrozumienia niniejszego podręcznika serwisowego.
- Niezastosowanie się do tego ostrzeżenia może spowodować urazy dostawcy usług, operatora lub pacjenta w wyniku porażenia elektrycznego, zagrożenia mechanicznego bądź innego.

ATENÇÃO

(PT-BR)

- Este manual de assistência técnica só se encontra disponível em inglês.
- Se qualquer outro serviço de assistência técnica, que não a gems, solicitar estes manuais noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- O não cumprimento deste aviso pode por em perigo a segurança do técnico, operador ou paciente devido a choques elétricos, mecânicos ou outros.

ATENÇÃO

(PT-PT)

- Este manual de assistência técnica só se encontra disponível em inglês.
- Se qualquer outro serviço de assistência técnica, que não a gems, solicitar estes manuais noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.

ATENȚIE

(RO)

- Acest manual de service este disponibil numai în limba engleză.
- Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere.
- Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service.
- Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.

ОСТОРОЖНО!

(RU)

- Данное руководство по обслуживанию предлагается только на английском языке.
- Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод.
- Перед обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения.
- Несоблюдение требований данного предупреждения может привести к тому, что специалист по обслуживанию, оператор или пациент получат удар электрическим током, механическую травму или другое повреждение.

UPOZORENJE

(SR)

- Ovo servisno uputstvo je dostupno samo na engleskom jeziku.
- Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilacke usluge.
- Ne pokušavajte da opravite uredaj ako niste procitali i razumeli ovo servisno uputstvo.
- Zanemarivanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehanickih i drugih opasnosti.

UPOZORNENIE

(SK)

- Tento návod na obsluhu je k dispozícii len v angličtine.
- Ak zákazníkovi poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka.
- Nepokúšajte sa o obsluhu zariadenia skôr, ako si neprečítate návod na obsluhu a neporozumiete mu.
- Zanedbanie tohto upozornenia môže vyústiť do zranenia poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, do mechanického alebo iného nebezpečenstva.

ATENCION

(ES)

- Este manual de servicio sólo existe en inglés.
- Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual.
- No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio.
- La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.

VARNING

(SV)

- Den här servicehandboken finns bara tillgänglig på engelska.
- Om en kunds servicetekniker har behov av ett annat språk än engelska ansvarar kunden för att tillhandahålla översättningstjänster.
- Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken.
- Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.

DIKKAT

(TR)

- Bu servis kilavuzunun sadece ingilizcesi mevcuttur.
- Eğer müşteri teknisyeni bu kilavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer.
- Servis kilavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz.
- Bu uyariya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "damage in shipment" written on all copies of the freight or express bill before delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

To file a report:

- Call 1-800-548-3366 and use option 8.
- Fill out a report on <http://us44hdd21/sctq/InstallFulfill/InstalFulfillment.htm>
- Contact your local service coordinator for more information on this process.

Rev. Jan. 5, 2005

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

LITHIUM BATTERY CAUTIONARY STATEMENTS

CAUTION



Risk of Explosion.

Danger of explosion if battery is incorrectly replaced.

Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the manufacturer's instructions.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

GE personnel, please use the Healthcare PQR Process to report all omissions, errors, and defects in this publication.

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Revision History

Revision	Date	Reason for change
1	10/23/08	Initial Release

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Preface

Publication Conventions

Section 1.0

Safety & Hazard Information

1.1 Text and Character Representation

Within this publication, different paragraph and character styles have been used to indicate potential hazards. Paragraph prefixes, such as hazard, caution, danger and warning, are used to identify important safety information. Text (Hazard) styles are applied to the paragraph contents that is applicable to each specific safety statement. Words describe the type of potential hazard that may be encountered and are placed immediately before the paragraph it modifies. Safety information will normally include:

- Type of potential Hazard
- Nature of potential injury
- Causative condition
- How to avoid or correct the causative condition

EXAMPLES OF HAZARD STATEMENTS USED

It is important that you read and understand hazard statements, and not just ignore them.



DANGER

DANGER IS USED WHEN A HAZARD EXISTS WHICH WILL CAUSE SEVERE PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- **ELECTROCUTION**
- **CRUSHING**
- **RADIATION**



WARNING

WARNING IS USED WHEN A HAZARD EXISTS WHICH COULD OR CAN CAUSE SERIOUS PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- **POTENTIAL FOR SHOCK**
- **EXPOSED WIRES**
- **FAILURE TO TAG AND LOCKOUT SYSTEM POWER COULD ALLOW FOR UN-COMMAND MOTION.**



CAUTION

Caution is used when a hazard exists which can or could cause minor injury to self or others if instructions are ignored. They include for example:

- **Loss of critical patient data**
- **Crush or pinch points**
- **Sharp objects**

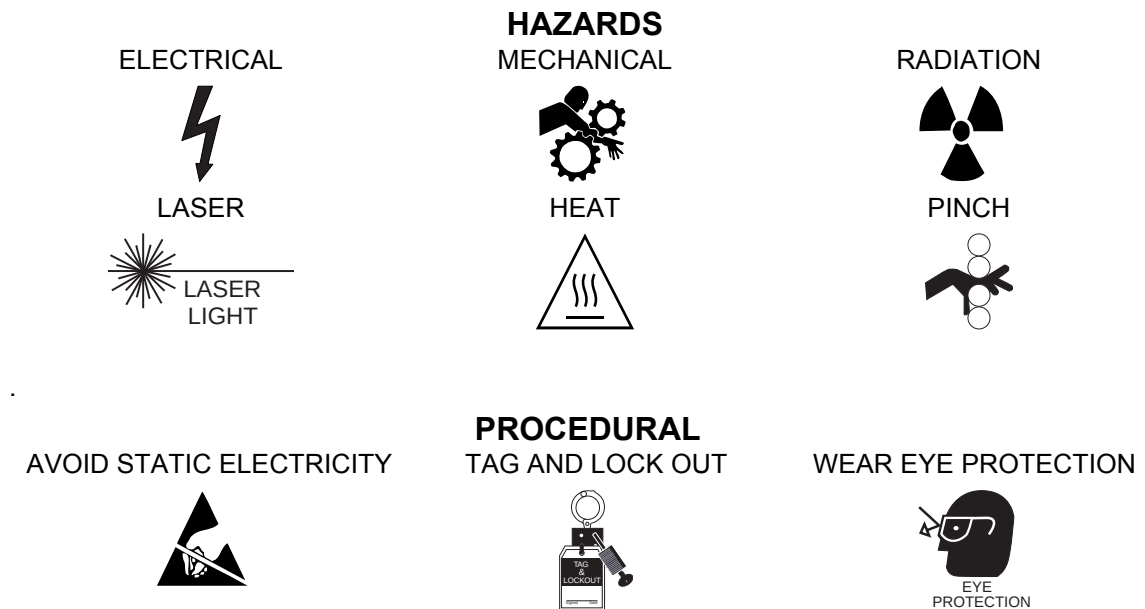


NOTICE Notice is used when a hazard is present that can cause property damage but has absolutely no personal injury risk. They can include:

- Disk drive will be destroyed
- Internal mechanical damage, such as to the x-ray tube
- Coasting the rotor through resonance.

1.2 Graphical Representation

Important information will always be preceded by the exclamation point  contained within a triangle, as seen throughout the Preface chapter. In addition to text, several different graphical icons (symbols) may be used to make you aware of specific types of hazards that could possibly cause harm.



Section 2.0 Buttons, Switches and Keyboard Inputs (Hard & Soft Keys)

Different character styles are used to indicate actions requiring the reader to press either a hard or soft button, switch or key. Physical hardware, such as buttons and switches, are called hard keys because they are hard wired or mechanical in nature. A keyboard or on/off switch would be a hard key. Software or computer generated buttons are called soft keys because they are software generated. Software driven menu buttons are an example of such keys. Soft and hard keys are represented differently in this publication.

Example: A power switch **ON/OFF** or a keyboard key like **ENTER** is indicated by applying a character style that uses both over and under-lined bold text that is bold. This is a hard key.

Example: Whereas the computer MENU button that you would click with your mouse or touch with your hand uses over and under-lined regular text. This is a soft key.

END OF SECTION

Chapter 1

Overview

Section 1.0

Hand-Held Controller

During needle biopsies, the Radiologist or Physician controls radiation with the foot switch, and uses the hand-held control (HHC) to view images on an in-room monitor. The HHC also provides alignment light, move-to-scan and cradle micro positioning controls. The Radiologist or Physician also has the option to release HHC control of the cradle, and use the cradle handle to manually position the patient in the scan field of view.

Refer to [Figure 1-1](#) and [Table 1-1](#), on page 20 for a description of each of the HHC push button functions.

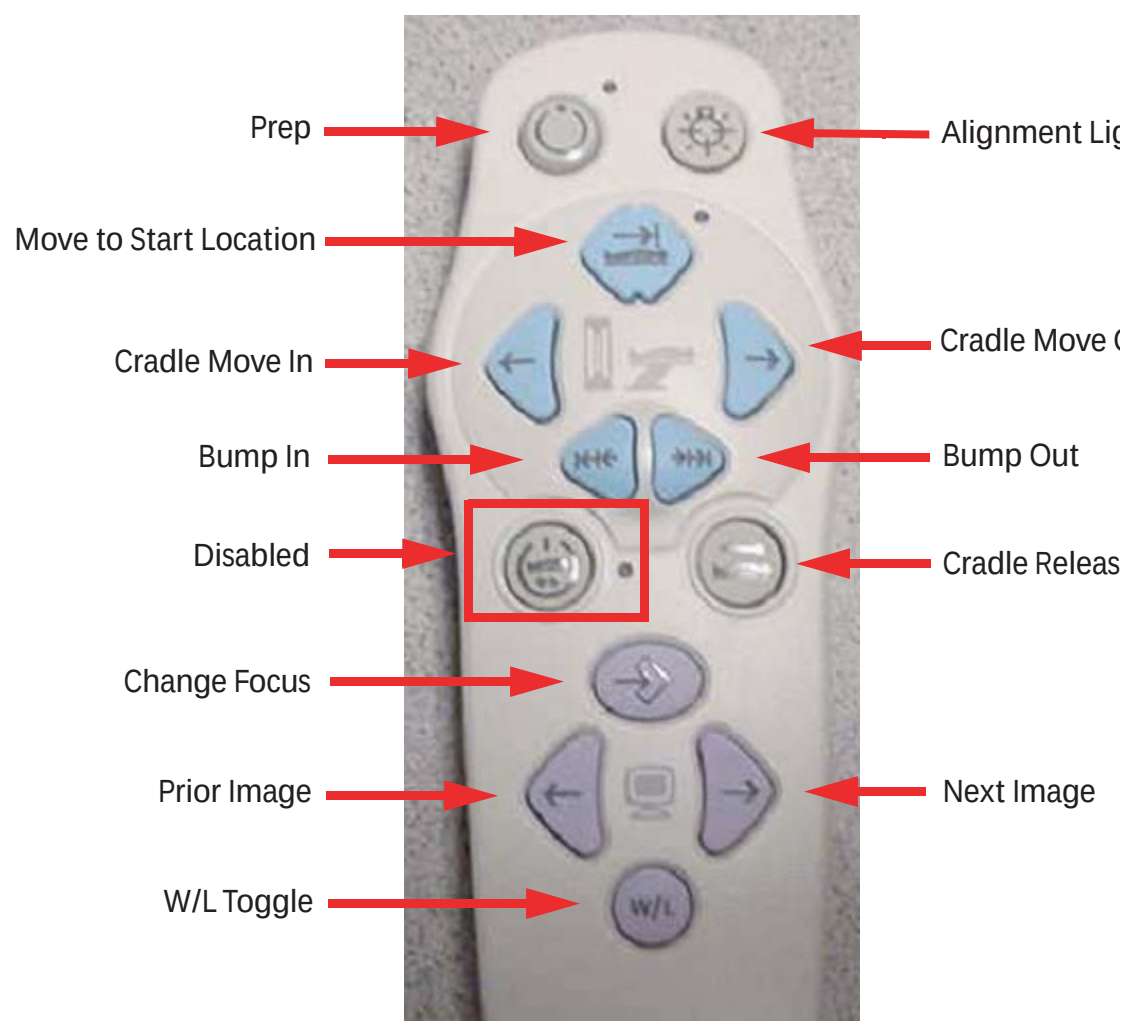


Figure 1-1 HHC Push buttons

Table 1-1 HHC Push button Functions

Button Name	Button	Description
Prep		Prepares the system for x-ray acquisitions.
Alignment Light		This button Enables the laser positioning lights.
Move to Start Location		This button positions the cradle to the start location prescribed by you in the SmartStep View/Edit screen.
Cradle Move In		The Cradle Move In button moves the cradle in towards the gantry.
Cradle Move Out		The Cradle Move In button moves the cradle away from the gantry.
Bump In		This moves the cradle a pre-defined bump distance towards the gantry. The bump distance is pre-defined on the View/Edit screen.
Bump Out		This moves the cradle a pre-defined bump distance away from the gantry. The bump distance is pre-defined on the View/Edit screen.
Disabled		This button is disabled. It is not used for SmartStep.
Cradle Release		This is a toggle button. It releases the cradle so it can be moved using the cradle handle. Pressing the button again latches the cradle.

Section 2.0

Prerequisites

- Must have all components in B7868SM Smart Step Collector - including software and hardware.
- Must have the correct hardware kit. (2376773, Rev. 2)
- Must have B7710LN hardware kit.

Chapter 2

Installation - Discovery CT750 HD

Section 1.0 Overview

The information in this Chapter applies to Discovery CT750 HD and systems with the new IPC.

The SmartStep hardware is shipped ready for immediate installation. There are two cables required for installation, 2355108 and 2286150. Locate these in the kit. See [Figure 2-1](#) for a general layout of cabling and connections.

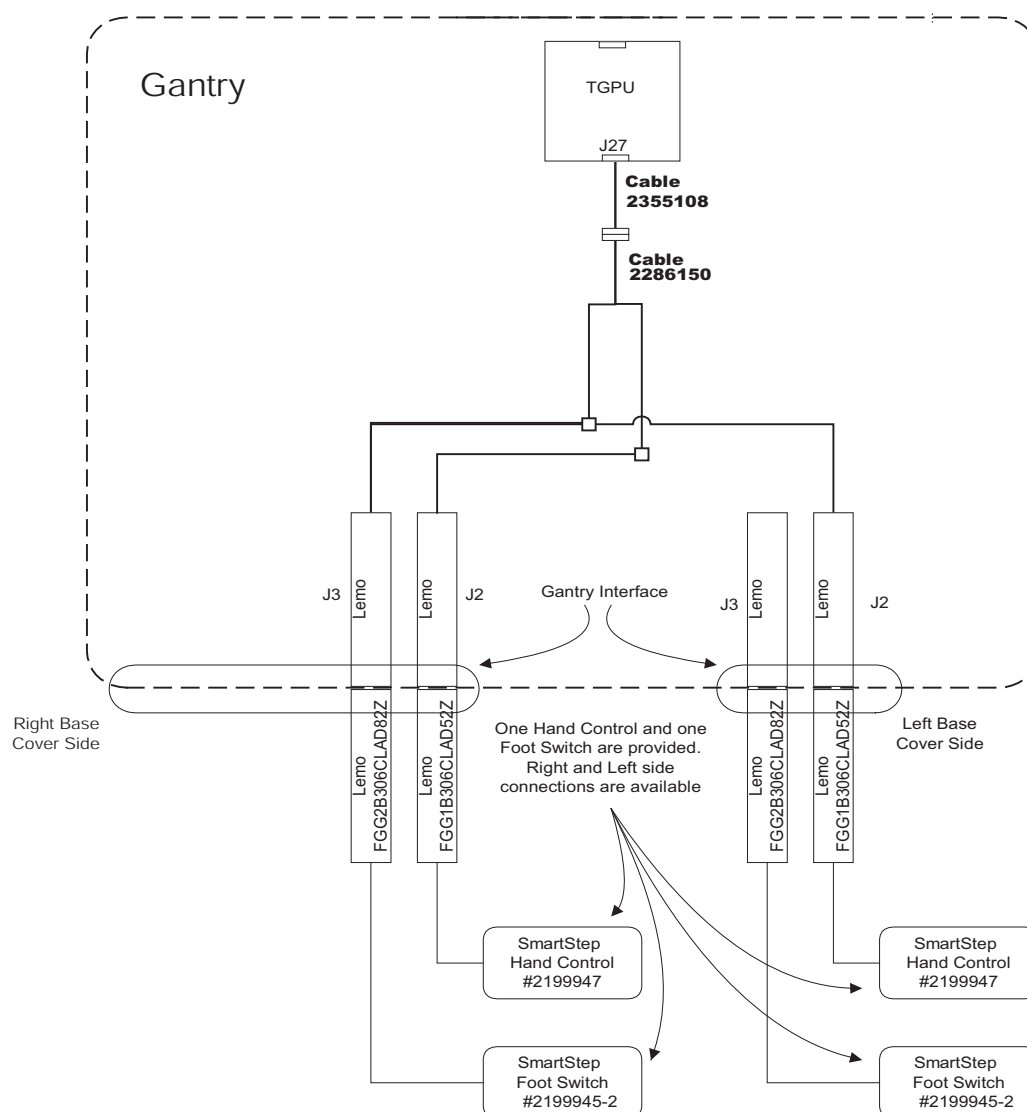


Figure 2-1 SmartStep Interconnect and Functional Drawing (HD)

Section 2.0 Procedure



DANGER RISK OF ELECTRIC SHOCK. USE PROPER LOCKOUT/TAGOUT PROCEDURES.

2.1 Preparation - Cover Removal

- 1.) Remove the two gantry side covers. Refer to your System Service Manual for any needed instruction.
- 2.) Turn **OFF** all three gantry service switches (Axial Drive, HVDC, 120 VAC).

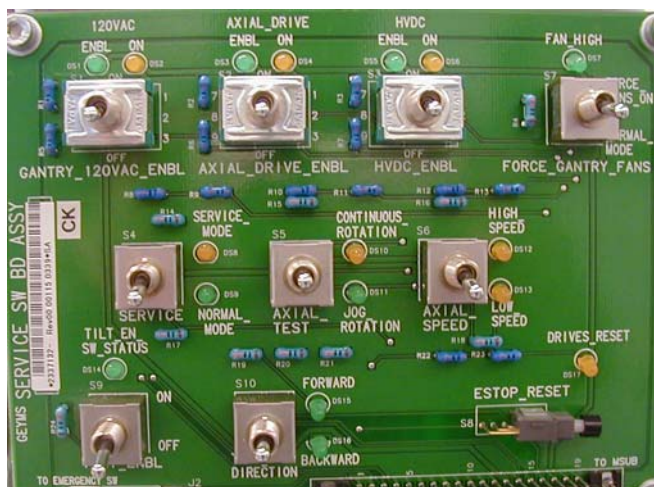


Figure 2-2 LightSpeed Gantry Service Switches

- 3.) Remove both left and right Gantry side base covers (items 6 & 7, respectively, in [Figure 2-3](#)) and then both rear base covers (items 8 & 9, in [Figure 2-3](#)).

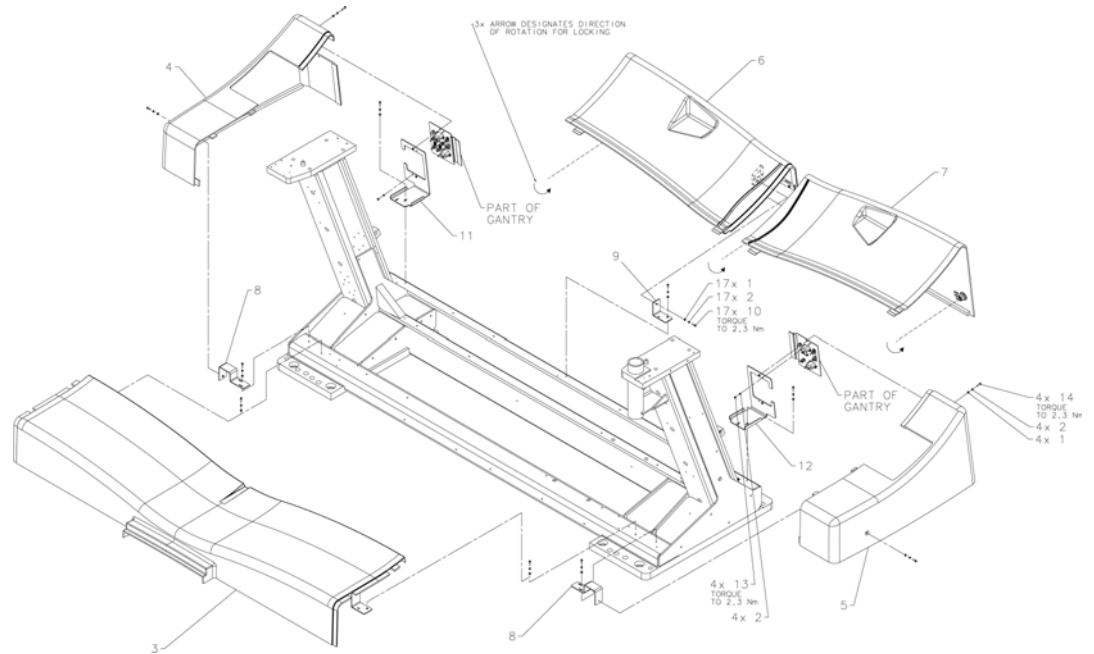


Figure 2-3 Gantry Base Covers, exploded view of an H2 base. Cover installation is same for gantry procedures.

Note: Do Not remove the front base cover! Typically only the two side and back covers need to be removed.

2.2 Installation - Faceplates

- 1.) Install the cable 2286150.
- 2.) Connect the LEMO plugs to the right and left option panels.
- 3.) Remove the plate on left side of option interface panel.
- 4.) With the outer conical nut removed, insert a LEMO connector through the appropriate hole in the faceplate, with the red dot facing UP (see [Figure 2-4](#)).

Top of Gantry. ↑



Figure 2-4 Orientation of Connectors

- 5.) Adjust the inside conical nut so enough thread appears through the hole and the external nut can be firmly attached while keeping the connection flush with the faceplate.
- 6.) Repeat steps 1 - 5 (above) for the right-side option interface panel.
- 7.) Route the cables and ty-wrap in place as shown in [Figure 2-5](#).



Route cable as shown along the gantry base's rear tube

Figure 2-5 SmartStep Cable Routing

2.3 Installation - Cables

- 1.) Locate the gantry MSUB. See [Figure 2-6](#).

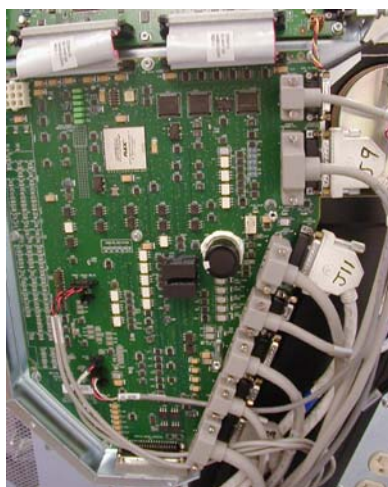


Figure 2-6 Gantry MSUB

- 2.) Locate J27. See [Figure 2-7](#).



Figure 2-7 Connector J27

- 3.) Install cable 2355108 to J27. See [Figure 2-8](#).

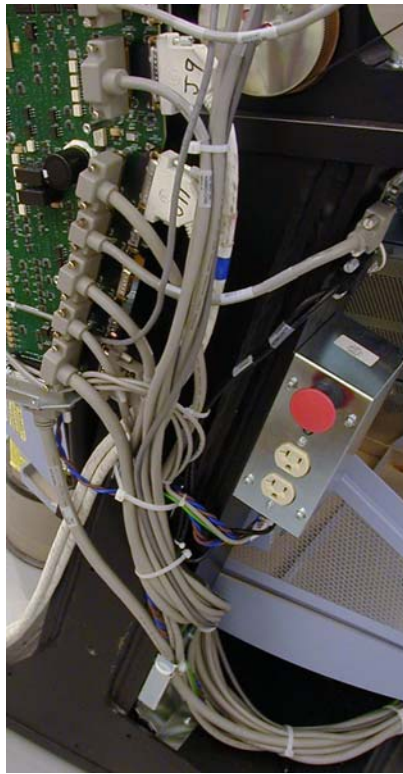


Figure 2-8 Install Cable 2355108

- 4.) Check that covers fit around the cables.

Chapter 3 Discovery PET CT 600/690

Section 1.0 Overview

The information in this section applies to the Discovery PET CT 600/690 with an IPC Panel. The SmartStep hardware is shipped already installed.

- 1.) Use the figure below to install the footswitch and handholder control.

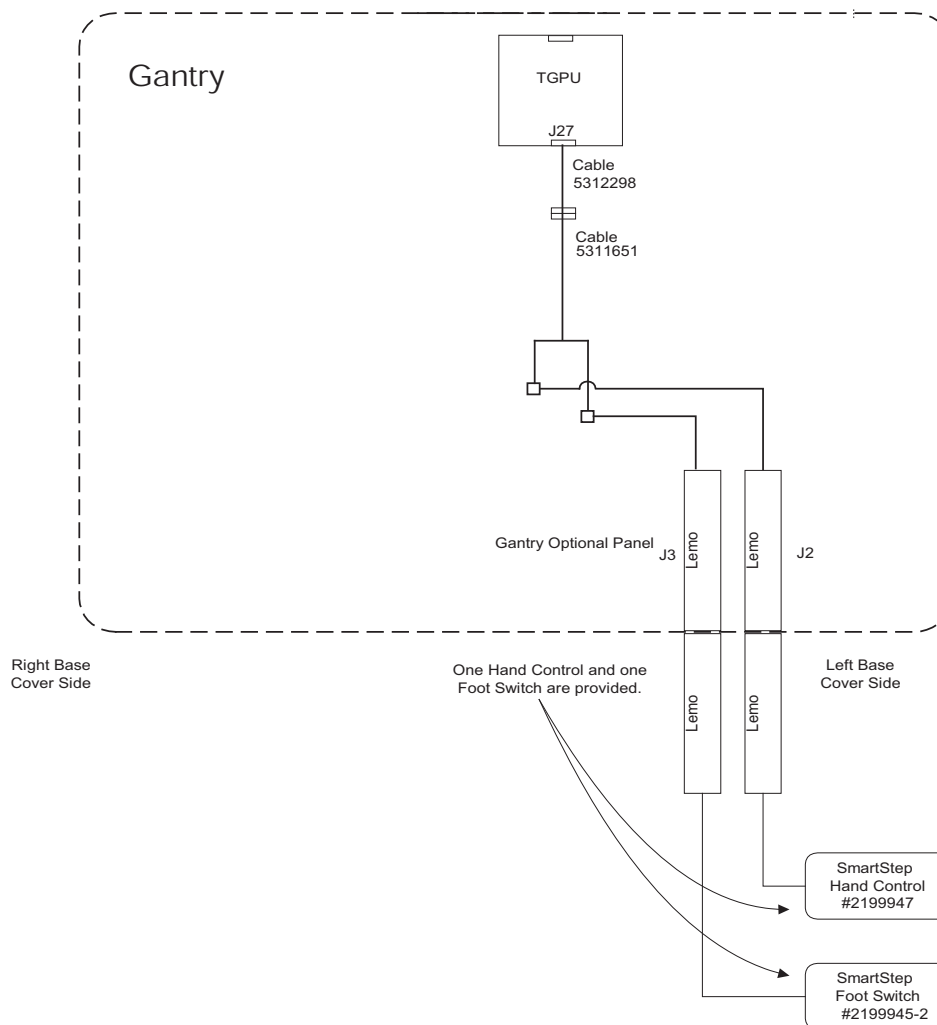


Figure 3-1 SmartStep Interconnect and Functional Drawing (PET)

- 2.) Confirm that the LEMO plugs are available in the IPC panel. If so, connect the footswitch and hand control assemblies.
- 3.) If the cable is not present, use the figure above to route the cable 2355108 and 228866150.
- 4.) Connect the end of cable 2355108 to the TGPU J27 and the other end to J15 of cable 228866150.
- 5.) Connect the LEMO connector to the IPC panel.
- 6.) Locate J27.



Figure 3-2 Gantry External Connections

Chapter 4

Patient Touch Leakage Test

Section 1.0 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	10 mins	20 mins	10 mins

Section 2.0 Overview

This procedure must be completed after all options have been installed. It covers three safety and leakage current checks:

- Patient Touch current test (completed after installation)
- System Ground resistance measurement (completed during installation)
- Ground current typical (completed after installation - optional)

Section 3.0 Preliminary Requirements

3.1 Tools and Test Equipment

Item	Qty
Standard service tool kit	1
Dale 600 Meter (from Tool Pool)*	1
Dale extended length leads*	1

*This procedure has been validated only with the Dale 600. GE Healthcare cannot guarantee the accuracy of this procedure, if any other meter is used.

3.2 Safety

- WARNING**  **POTENTIAL FOR SHOCK**
GROUND WIRES WILL HAVE GROUND CURRENT PRESENT WITH POWER “ON”. FOLLOW APPROPRIATE SAFETY PROCEDURES FOR WORKING WITH AN ENERGIZED SYSTEM.
- NOTICE** Follow ALL required safety and PPE procedures customary for your organization, when working on this product.

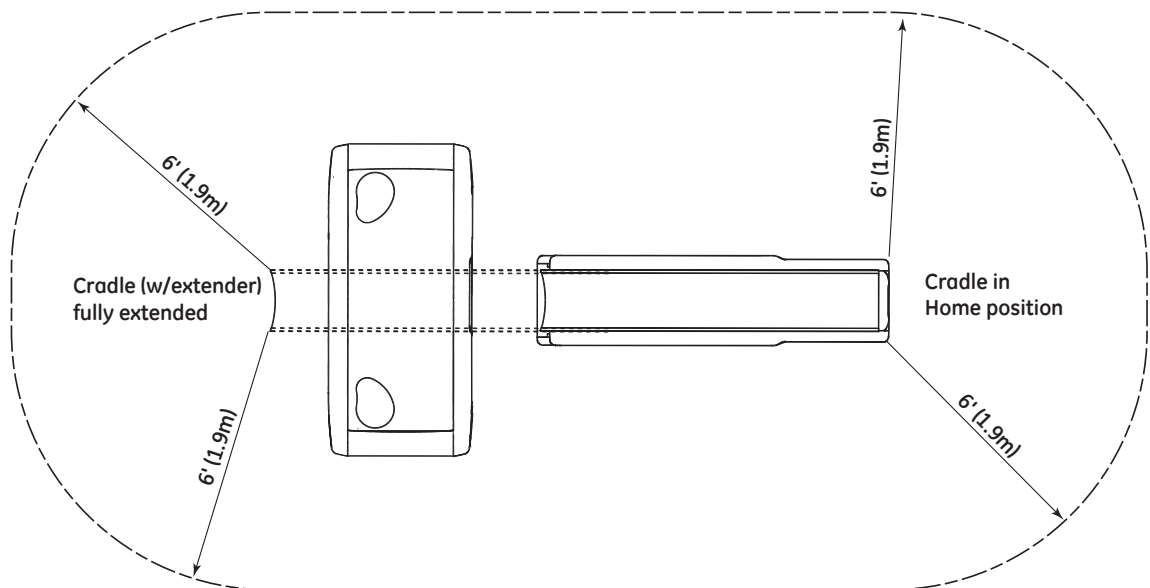
3.3 Required Conditions

- Only trained service personnel should service the GE CT Scanner.
- Footswitch cover must be removed.

Section 4.0 Procedure

4.1 Test Conditions

- Test with Table at maximum elevation and again with Table at minimum elevation
- Test to cover all points in an envelope described by Table travel from minimum to maximum extension (including table extender).
- Test from both the head and the foot of the Table -- e.g., test assuming patient feet first position as well as patient head first
- Test above (7ft or 2.1m) and below (6ft or 1.9m) Table surface -- e.g., test assuming patient lying face down as well as face up.
- Test for six feet (1.9m) envelope preventing patient access to all conductive surfaces including:
 - IV poles and tray assembly
 - any smart step monitors and stands
 - table bearing rails
 - all other conductive surfaces



4.2 Patient Touch Leakage Current Test Procedure

NOTICE These instructions have been validated only with a Dale 600/601 meter. Due to the unique nature of this meter, GE Healthcare cannot guarantee the accuracy of these procedures when performed with any meter other than the Dale 600/601.

- 1.) Move the table to ISO elevation.
- 2.) Remove the footswitch covers and the gantry left side cover.
- 3.) Refer to the Dale 601 Operator's Manual for instructions on the use of the Dale 600 meter for measuring leakage current (or refer to [Figure 4-1](#), for a quick overview).

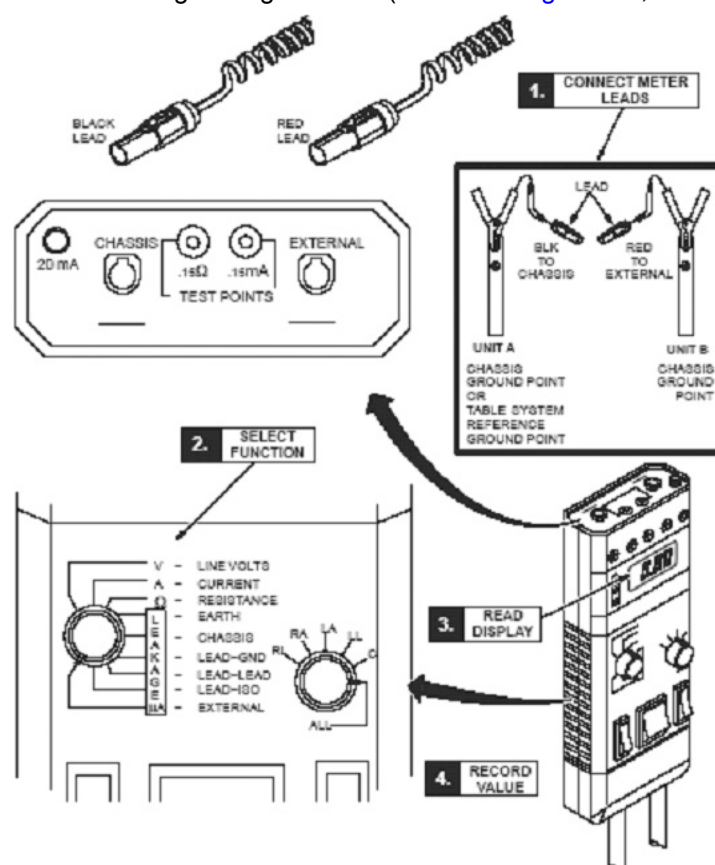


Figure 4-1 Using Dale 600 to Measure Leakage Current

- 4.) Plug the Dale 600 Meter into the outlet on the gantry left side. Confirm the outlet is correctly wired per the three LED indicators on the meter.
- 5.) Connect one end of the shorter black lead to the chassis plug and connect the other end to the table ground bus.
- 6.) Connect the longer red test lead (or the longer black lead) to the external plug on the top of the Dale 600 meter.
- 7.) Set the function switch on the Dale to external. Use the external lead to touch the meter's test terminal, to test that the meter is operational.
- 8.) The black lead will be connected to the table base ground bus, and the read lead will be connected to the devices (components) under test.

Note: Your meter may have two black leads that are keyed for chassis and the ground connection. A valid calibration sticker must be present on the meter used. Record this information on the GE 4879 form.

- 9.) Beware: Leakage current cannot exceed the following and is tested with power ON:
- a.) Critical care areas (invasive) - 10µa CT Systems
 - b.) General care areas - 20µa**
 - c.) Not intended for patient area - 50µa
- 10.) Testing must be completed between the system reference ground point (table base) to unit reference ground points (i.e. gantry and table, see [Figure 4-3](#)).
- 11.) Test all conductive surfaces and components within patient reach or within 6 feet (1.9m) of the table and 7 feet (2.1m) above.
- Measure at table maximum travels.
 - At some sites, wall outlet cover plates and sinks may become an issue.
- 12.) Test all Optional components such as in-room monitors, injector, overhead monitor suspensions, and table options.
- Record these results in the data sheet. To be completed at install and during PM inspections.

Gnd Bus to	Install <10µA
Any table ground points within the 6' range	
Any gantry ground points within the 6' range	
Injector assembly metal surface	
Boom-in-Room metal surface	
Monitors or metal surface	
Sink or metal surface	
Installed table accessories	
(other)	
(other)	
(other)	

NOTICE

Be Aware of Static Discharge from: scan window, keypads, display, and touch pads or other plastic surfaces.

- 13.) **Procedure Hints:**
- a.) Look for items that have abnormal measurements, high or low.
These could indicate mis-wiring, loose connections, or poor connections due to corrosion, painted surfaces, etc.
 - b.) High leakage could indicate a wiring error such as a neutral connected to the ground.
 - c.) Fluctuating ground currents could indicate a short, poor connection, or facilities ground problem causing leakage currents from other areas of the facility to flow through our system grounds.

d.) Refer to the illustrations below, for measurement visual descriptions.

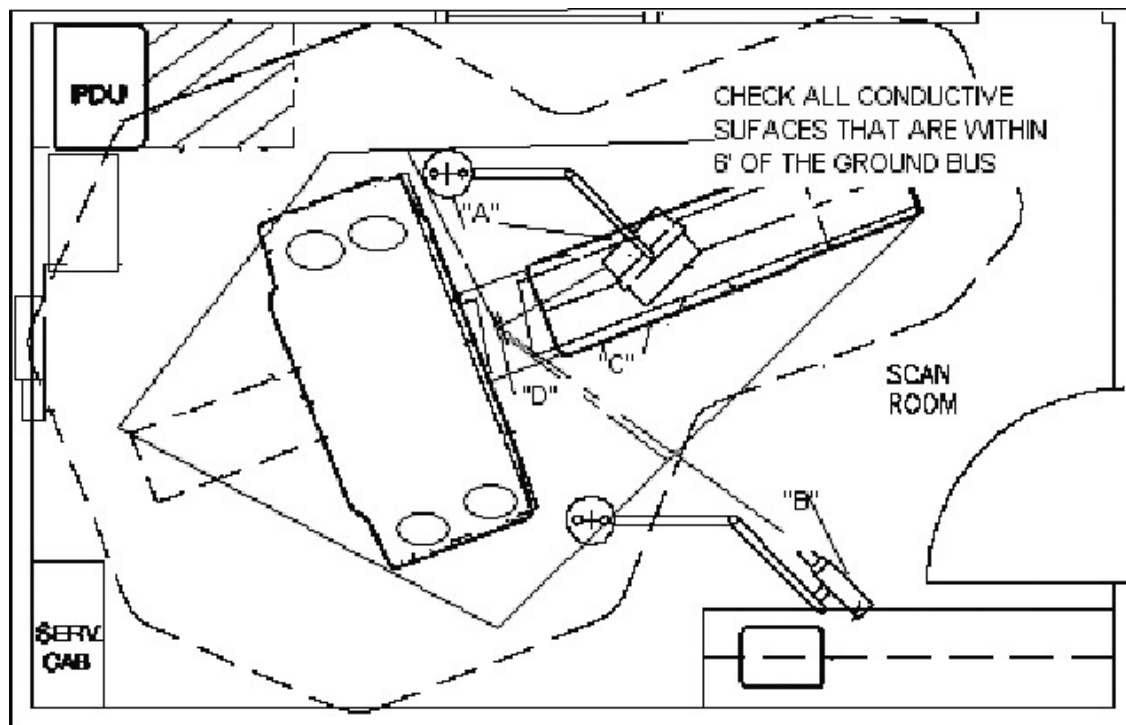


Figure 4-2 CT Leakage Current Map

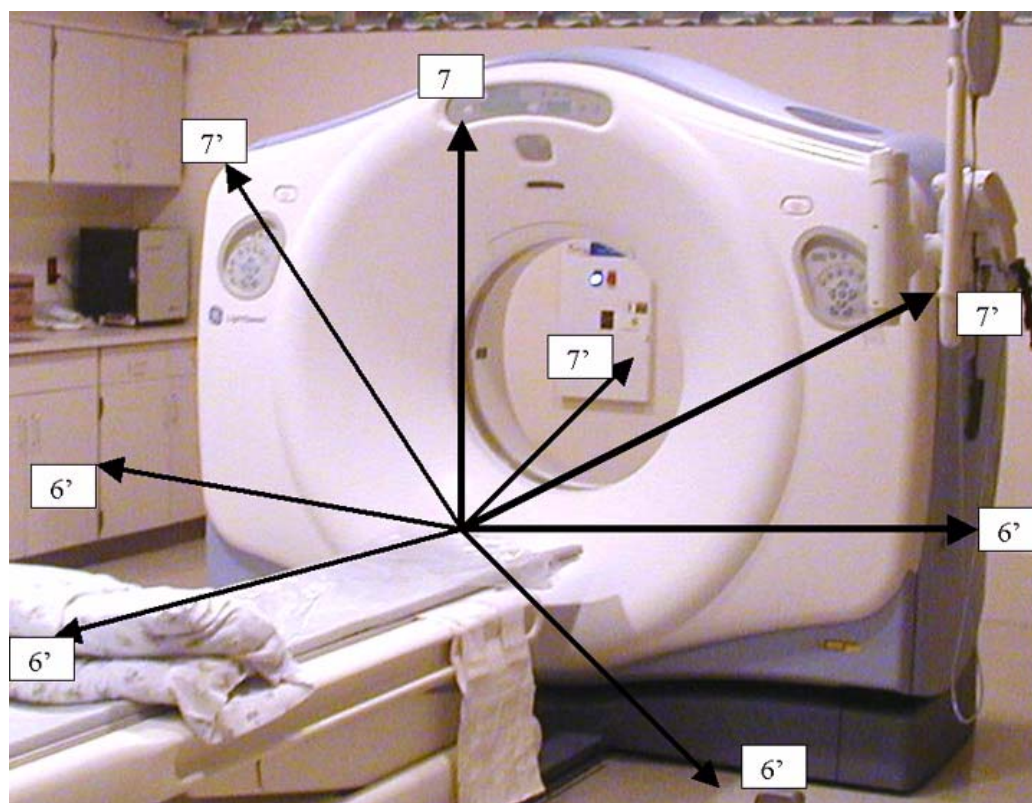


Figure 4-3 Measurement Points

4.3 Ground Resistance Checks Procedure (done during mechanical install)

Use Fluke 87 to measure ground resistance.

- 1.) Use a Fluke 87 or equivalent RMS Meter to verify the presence of less than 0.5 ohm of resistance between each of the following points:

FROM	TO	
PDU Ground Bus	Vault Ground	☐ Check box when complete
PDU Ground Bus	Table/Gantry raceway ground point	☐ Check box when complete
Table/Gantry raceway ground point	Gantry Chassis	☐ Check box when complete
Table/Gantry raceway ground point	Table Chassis	☐ Check box when complete
Table/Gantry raceway ground point	Operator Console Chassis	☐ Check box when complete
All Display or Computing Options (if any)	Operator Console Chassis	☐ Check box when complete

Table 4-1 Resistance Verification - Site Ground

- 2.) Connect the meter leads to the Fluke 87 meter.
- 3.) Set the function switch to resistance ohms.
- 4.) The black and red leads will be connected per Table 4-1.
- 5.) All ground resistance testing is done with power OFF. Ground current will cause errors in the resistance readings.
- 6.) Use lockout/tag out procedures.
- 7.) Maximum ground resistance is 0.5 ohms impedance.
- 8.) Using the table footswitch ground bar as a reference. Check the ground resistance to all 5 system grounds.

Record the measurements in the datasheet:

SYSTEM GROUND BUS	TYPICAL READING	SITE DATA
Gantry Ground Stud	0.5 Ohms	
Console Ground Stud	0.5 Ohms	
PDU Ground Stud	0.5 Ohms	
Table Ground Stud	0.5 Ohms	

Hints:

- Look for undersized ground wires and repair.
- Ground resistance readings should not fluctuate. If so this indicates an intermittent

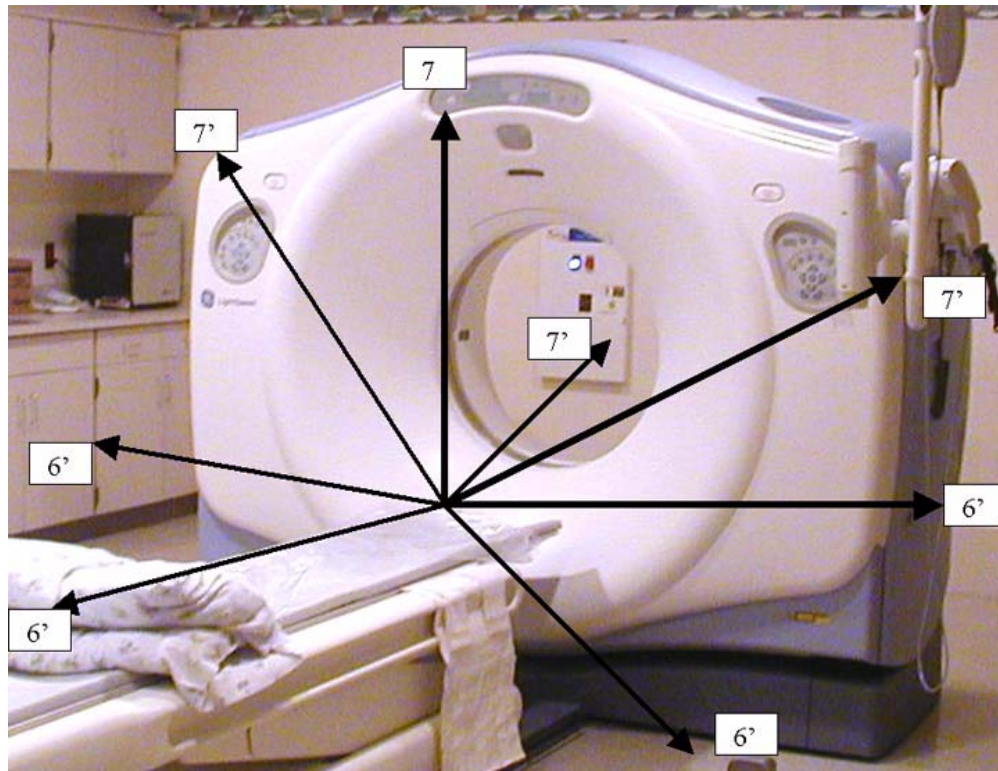


Figure 4-4 Measurement Points

Chapter 5

Finalization (All Systems)

Section 1.0

Installation - SmartStep Cradle Handle and HHC Holder

- 1.) Refer to [Figure 5-1](#).

Press the Cradle Release/Engage button to release HHC control.
Then, use the Cradle Handle to manually position the patient.

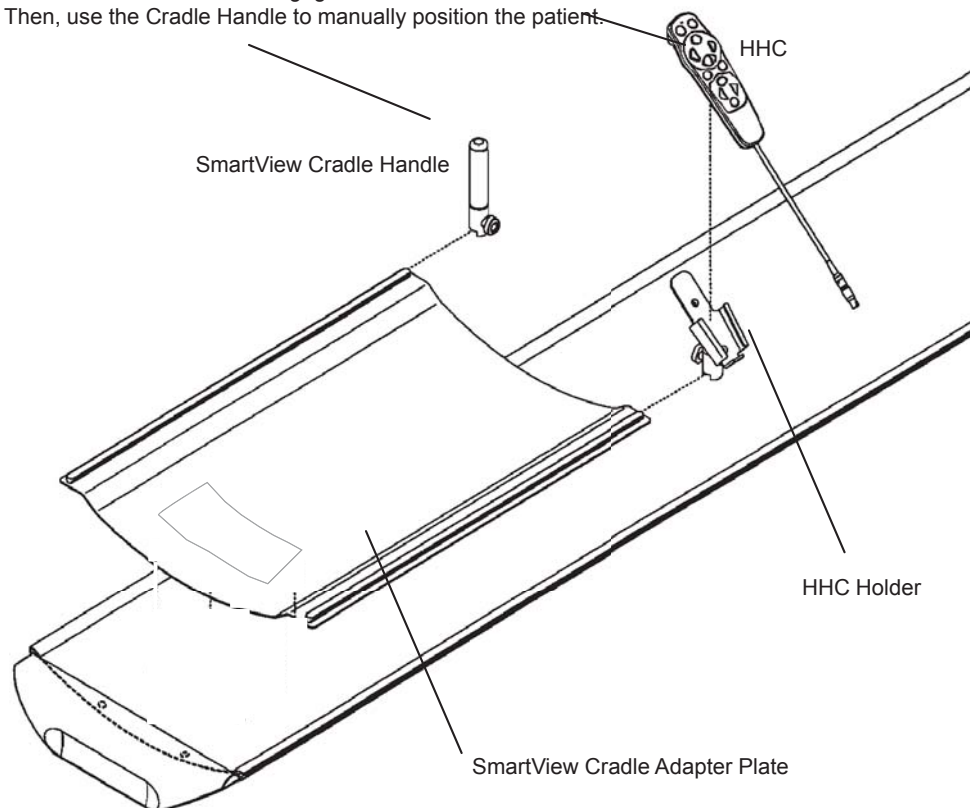


Figure 5-1 SmartStep Cradle Handle and HHC Installation

- 2.) Remove the patient pad from the cradle and set aside.
- 3.) Orient the cradle handle adaptor plate, and lay it on the cradle top.
- 4.) Return the patient pad to the cradle.
- 5.) Slide the HHC holder and cradle handle onto the cradle adapter plate.
- 6.) Tighten the knurled knobs to hold in place.
- 7.) Connect the foot switch and hand holder to the gantry interface options panel.

Note:

Section 2.0 Installation - Console Modifications

- 1.) Remove the modification plate from the upgrade kit.
- 2.) Refer to [Figure 5-2](#).
- 3.) Attach the new plate to the left of the existing rating plate(s).
- 4.) Take care not to cover any part of the existing plate(s). Install rating plate on console.
- 5.) Referring to your System Software Load From Cold Manual, install the SmartStep Option Key.



Figure 5-2 GOC5 modification plate

Section 3.0 Finishing Up - Testing & Cover Re-Installation

- 1.) Be certain all obstructions to gantry motion are clear and turn **ON** all 3 switches (Axial Drive, HVDC, 120 VAC) on the service switch panel.
- 2.) Run a system functional check from the Service CD located under Function Checks, including a test scan, before installing the covers.
- 3.) Check SmartStep Operation - refer to SmartStep section of the Operator's Manual for procedure.
- 4.) Turn **OFF** all 3 switches (Axial Drive, HVDC, 120 VAC) on the service switch panel.
- 5.) Reinstall both rear base covers (See [Figure 2-3, on page 25](#) or [Figure 2-3, on page 25](#)).
- 6.) Reinstall the gantry side covers. Refer to your System Service Manual for any needed instruction.
- 7.) Perform a test scan to confirm operation.

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